Projection before Diophantine Encoding

A smaller direct quartic whose insolubility is equivalent to Con(ZFC)

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Vladimir Reshetnikov's ProveIt project

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Abstract

We construct a completely specified polynomial in 7,617 unrestricted integer variables, of exact total degree four, whose lack of integer solutions is provably equivalent in ZFC to syntactic consistency of ZFC. Its complete collected coefficient list and quadratic residuals are supplied. It has 148,220 nonzero monomials and largest absolute coefficient 8 974 652 400 619 654 748 160.

The construction starts directly from the twelve numerical C222 proof rules. It does not simulate a Turing machine or invoke Jones's universal polynomial. Relative to the supplied direct quartic in 27,916 integer variables, the new equation removes 20,299 variables (72.71 percent). The mechanism is to eliminate computed formula codes before encoding an unbounded family of proof steps, then pad proofs by a zero-balance truth rule so that proof length bounds the remaining rule parameters. After the additional coprimality and remainder-bound refinements, the local family needs eight witness columns rather than twenty-nine. This cuts the number of exponentiation certificates from 108 to 40. Signed arithmetic and signed congruence quotients reduce the cost of the final quartic conversion.

This is a substantial simplification of the direct, degree-four branch. It is not a minimum-variable record: the repository already contains nine-positive-variable and ten-integer-variable equations of much higher degree. We give ordinary mathematical proofs, exact data and algebra checks, and reproducible builders. The logical calibration imports the repository's C222-to-ZF and ZF-to-ZFC results. A new end-to-end Lean or encoded object-language ZFC derivation is not supplied.

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1 Result, scope, and exact equation

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$; the final equation is over $\mathbb{Z}$, without sign restrictions on any quantified coordinate. The phrase "simplest equation" has no unique mathematical ordering: number of variables, degree, coefficient height, arithmetic circuit size, and description length can move in different directions. We therefore identify the comparator and give literal measurements.

The starting point is the fully materialized direct quartic in the incoming package *Direct and centered Diophantine consistency equations*, hereafter Round12 [1]. The repository snapshot also contains later arithmetic reductions not summarized by the historical collection README. We inspected both that collection and the active research, including the new thirteen-integer radical-norm package. The latter is not the source of the equation in this paper.

Construction	Integer variables	Degree	Monomials
Round12 direct baseline	27,916	4	758,275
Signed quotients on that baseline	23,048	4	717,327
This paper's completed equation	7,617	4	148,220

Table 1: A comparison within the direct proof-rule construction. All variables in this table are unrestricted integers.

Definition 1.1 (The actual polynomial)**.** Let $r_1, \ldots, r_{4,379}$ be the literal quadratic polynomials in `equations/residuals_integer.json`, in its stored order. Define

$$P_{\text{direct}}(x_0, \ldots, x_{7,616}) = \sum_{\ell=1}^{4,379} r_\ell(x_0, \ldots, x_{7,616})^2 = 0. \tag{1.1}$$

Its expanded coefficient list is in `equations/equation_integer.json`. A term `[c,[i1,...,ik]]` means $c x_{i_1} \cdots x_{i_k}$; indices start at zero, repetitions are powers, and an empty index list denotes the constant term. Each coefficient is an explicit decimal integer. The file contains neither an unevaluated oracle nor a variable exponent, factorial, division, square root, or bounded quantifier.

The sum notation in (1.1) is a fixed finite abbreviation. The JSON is a convenient exact representation of an ordinary polynomial, just as a long coefficient table would be. The builders specify every entry; an independent checker reconstructs the collected sum and compares it exactly. Definitions of intermediate expressions do not add hidden quantifiers to this displayed variable count. Every retained multiplication wire and every natural-number representation witness is counted.

Theorem 1.2 (Main ordinary theorem)**.** *For the particular polynomial* (1.1)*, with the fixed proof calculus and the inherited logical calibration described below,*

$$\text{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{7,617} P_{\text{direct}}(\boldsymbol{x}) = 0\right) \longleftrightarrow \text{Con}(\text{ZFC}). \tag{1.2}$$

The total degree is exactly four.

Here $\text{Con}(\text{ZFC})$ is the usual arithmetical assertion that there is no finite syntactic ZFC refutation. No consistency assumption is used to prove the biconditional. Its proof is completed in Section 12. It neither proves the left side nor proves consistency. Global minimality and historical priority of the general compiler ideas are not asserted.

2 The direct proof calculus and its calibration

The source is a fixed finite list of twelve single-conclusion numerical rule schemata. Formula constructors are integer polynomials, beginning with four times the Cantor pairing function:

$$\begin{aligned} \mathsf{I}(a,b) &= 2(a+b)(a+b+1)+4b, \\ \mathsf{R}(a,b) &= \mathsf{I}(a,b)+1, \qquad \mathsf{A}(a,b) = \mathsf{I}(a,b)+2, \qquad \mathsf{E}(a,b) = \mathsf{I}(a,b)+3, \\ \mathsf{N}(a) &= \mathsf{I}(a,1), \qquad \mathsf{X}(a,b) = \mathsf{N}(\mathsf{A}(a,\mathsf{N}(b))). \end{aligned} \tag{2.1}$$

The truth code is zero and the refutation target is one. The collision $\mathsf{I}(0,0) = 0$ is already covered by the inherited calculus semantics. The complete rules are specified in Appendix A and the frozen source data. Their four shared base parameters are $a, b, c, d \in \mathbb{N}$. Unused parameters may be zero.

Each rule has two padded premise labels p, q and one conclusion label v. An axiom uses $p = q = 0$. Generalization has one zero premise. The truth rule has $(p, q, v) = (0, 0, 0)$, so it has no contribution to any positive-label balance. Only the bundled axiom has an extra side condition. It is encoded exactly by

$$\big((c-a)(c-b)\big)^2 - (d+1)^2 = 0, \tag{2.2}$$

which, for natural parameters, says $c \neq a$ and $c \neq b$. Indeed a nonzero integer product has a positive absolute value, and one may choose $d = |(c-a)(c-b)| - 1$. Both signs of the differences are allowed.

The logical facts imported from the repository are

$$\begin{aligned} \mathcal{C}_{222}(1) &\longleftrightarrow \mathrm{Ref}_{\mathrm{ZF}}, \\ \mathrm{Ref}_{\mathrm{ZF}} &\longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\,. \end{aligned} \tag{2.3}$$

The first is the mathematical content of `KEndpoint.c222_iff_zf_refutation`; the second is `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. These declarations and their source files are pinned in the archive. The second equivalence uses the constructible-universe consistency transport. The first uses the fixed numerical calculus, its completeness and the set-theoretic seed calibration. Neither equivalence requires the execution of K215 in this construction [2, 3].

These are inherited mathematical theorems, not newly proved claims about the Python compiler. Their ordinary finite coding, completeness and constructibility arguments can be performed in ZFC. Referring to an external Lean proposition by itself would not establish the existence of an encoded object-language ZFC derivation. We keep that distinction throughout the paper.

3 Finite proof balances and one obligation accumulator

We recall the direct baseline's proof certificate, since the padding reduction depends on its exact structure. For each rule instance, associate the vector

$$e_v - e_p - e_q,$$

where e_0 is interpreted as zero and e_j, $j > 0$, is the unit vector at positive label j.

Lemma 3.1 (Single-conclusion balance certificate)**.** *Target one is derivable if and only if a finite multiset of valid rule instances has balance e_1.*

Proof. Unfold a finite derivation into a proof tree. Each internal premise occurrence cancels the corresponding conclusion occurrence of its subproof; only the root target remains. Truth leaves contribute zero.

Conversely, take a finite multiset with balance e_1. Let D be the least set of labels derivable using just its finitely supported rule instances, with the zero label regarded as available. Suppose $1 \notin D$. Sum the balance over the positive labels outside D. A rule whose conclusion lies outside D has at least one premise outside D; otherwise closure would put its conclusion in D. It therefore contributes at most zero. A rule with conclusion in D also contributes at most zero. But the required target vector contributes one, a contradiction. This argument permits cycles, repeated premises, and self-loops. It needs nonnegative rule multiplicities and one conclusion per rule. □

Suppose a list has length N and all its labels are at most h. Define

$$\begin{aligned} A(X) &= \prod_{i<N}(1+v_iX), \\ B(X) &= (1+X)\prod_{i<N}(1+p_iX)(1+q_iX). \end{aligned} \tag{3.1}$$

Every coefficient of $A - B$ has absolute value at most $2(h+1)^{2N}$. Since $h+1 \le 2^h$ for $h \in \mathbb{N}$, the base

$$M = M_0 + 1, \qquad M_0 = 2^{2hN+1} \tag{3.2}$$

strictly exceeds this coefficient bound.

Lemma 3.2 (Integer evaluation recovers the balance)**.** *If $A(M) = B(M)$ with M as above, then $A = B$, and the positive-label balances equal e_1.*

Proof. If a nonzero integer polynomial with coefficient magnitudes below M vanishes at M, let j be its least nonzero coefficient index. After dividing by M^j and reducing modulo M, that coefficient is a nonzero multiple of M, impossible. Hence $A = B$. For each positive integer r, compare multiplicities of the distinct rational root $-1/r$. The constant factors coming from zero labels do not contribute. The extra $1 + X$ contributes the target occurrence. □

Use one accumulator $S_0, \ldots, S_N$ with

$$\begin{aligned} S_0 = M+1, \qquad S_N = 1, \\ S_{i+1}(1+v_iM) = S_i(1+p_iM)(1+q_iM). \end{aligned} \tag{3.3}$$

All linear factors are positive. Starting from $S_0 > 0$, every natural accumulator value satisfying an update is positive, and telescoping gives $A(M) = B(M)$. Conversely, expand a finite proof tree by its open obligations: replace a target obligation by its premises. Take S_i to be the product of the current obligation weights $1 + rM$. This gives the required integral accumulator with no rational intermediate values. Inserting a truth step leaves S_i unchanged.

4 Eliminate branch-local data before encoding

Round12's local row quantified three labels, four base parameters, thirteen constructor wires, six beta coordinates, and three label-bound slacks: twenty-nine natural witness columns. The large cost is that each column receives its own binomial certificate in the bounded-quantifier elimination. Eliminating it only at the final quartic stage misses most of the possible saving.

Lemma 4.1 (Projection through a finite disjunction)**.** *Let a finite collection of branches be specified by polynomial equations, with additional coordinates $\boldsymbol{u}$ that are uniquely computed by a topologically ordered polynomial graph $\boldsymbol{u} = \boldsymbol{f}_j(\boldsymbol{a})$ in branch j. Assume these graph values lie in the required domain whenever $\boldsymbol{a}$ does. Then*

$$\exists \boldsymbol{u} \bigvee_j \big(\boldsymbol{u} = \boldsymbol{f}_j(\boldsymbol{a}) \wedge R_j(\boldsymbol{a}, \boldsymbol{u}) = 0\big) \quad \longleftrightarrow \quad \bigvee_j R_j(\boldsymbol{a}, \boldsymbol{f}_j(\boldsymbol{a})) = 0.$$

The branches may share the same scratch-coordinate names and have different graph assignments.

Proof. A satisfying assignment selects one branch and hence its graph values. Conversely, choose the graph values of a satisfying projected branch. Other branches need not vanish. A finite product of sums of squares implements precisely this disjunction over either $\mathbb{N}$ or $\mathbb{Z}$. □

In C222 every constructor wire is a polynomial with nonnegative coefficients. The exposed labels are computed as follows. Axioms have two zero premises and a conclusion given by their constructor expression; modus ponens has $(p, q, v) = (a, \mathsf{I}(a, b), b)$; generalization has $(p, q, v) = (b, 0, \mathsf{A}(a, b))$; truth has three zero labels. Thus all sixteen labels and wires can be substituted inside their selected branch. The bundled side condition (2.2) is retained, including its natural parameter d. It is not a deterministic constructor.

Write the resulting branch labels as $p_j(\boldsymbol{a}), q_j(\boldsymbol{a}), v_j(\boldsymbol{a})$, where $\boldsymbol{a} = (a, b, c, d)$. They are nonnegative-coefficient polynomials. Let Γ_j be zero except in the bundle branch, where it is the square of (2.2). If S, S' are the adjacent accumulator values, set

$$D_j = \Gamma_j + \big[S'(1 + v_j M) - S(1 + p_j M)(1 + q_j M)\big]^2. \tag{4.1}$$

Each D_j is nonnegative, and $\prod_{j=0}^{11} D_j = 0$ chooses a valid rule and its accumulator update. The update must be inside the branch after label substitution: making a single global substitution for labels shared by incompatible branches would be incorrect.

This substitution can greatly increase the degree of a local expression. It does not increase the final degree beyond four, because the local expression is retained as a shared arithmetic graph and is flattened only after the bounded family has been eliminated. It exchanges cheap deterministic multiplication wires for expensive encoded witness columns.

5 Padding replaces a separate label bound

Define the following fixed nonnegative-coefficient polynomial:

$$H(N) = \sum_{j=0}^{11} \big(p_j(N, N, N, N) + q_j(N, N, N, N) + v_j(N, N, N, N)\big). \tag{5.1}$$

The invalidity of the bundle side condition on the diagonal does not matter: H is a majorant for its label polynomial, not a purported valid diagonal instance. If $0 \le a, b, c, d \le N$, monotonicity gives $p_j, q_j, v_j \le H(N)$ for every branch.

Theorem 5.1 (Proof-length padding)**.** *C222 derives one if and only if there is an accumulator certificate of length N in which every rule's four base parameters are at most N, and the base is*

$$M_0 = 2^{2H(N)N+1}, \qquad M = M_0 + 1. \tag{5.2}$$

Proof. For completeness take a finite proof tree and its finite sequence of expansion rules. Choose N at least the sequence length and at least every base parameter occurring in those rules. This includes the bundle's d, chosen from its nonzero product. Pad to exactly N steps by truth rules with all four parameters zero. Every label is now at most $H(N)$. Choose M using (5.2), and only then construct the accumulator from open obligations as in Section 3. Padding has no effect on balances or products.

For soundness, monotonicity of H bounds every label. The coefficient-height argument applies with $h = H(N)$. The accumulator therefore gives a balance certificate, and Lemma 3.1 yields a derivation. A zero-length list is automatically rejected: its endpoints would require both $S_0 = M + 1$ and $S_0 = 1$, while $M \geq 3$. □

The order of choices is essential:

finite proof parameters $\longrightarrow$ N $\longrightarrow$ M $\longrightarrow$ (S_i) $\longrightarrow$ beta codes and their witness bound U.

There is no circular requirement that M dominate a bound which already contains its own accumulator values. The separate label bound m and its three per-row slacks disappear. This is why four rule parameters may use bound N, while six beta coordinates use a different bound U.

6 Exactly ten local witness columns

Encode the single finite accumulator with the ordinary beta function

$$\beta(x, y, i) = x \bmod (1 + (i + 1)y).$$

For fixed index i, use the triples (S, q_0, s_0) and (S', q_1, s_1) to assert

$$\begin{aligned} x &= q_0(1 + (i + 1)y) + S, & 1 + (i + 1)y &= S + s_0 + 1, \\ x &= q_1(1 + (i + 2)y) + S', & 1 + (i + 2)y &= S' + s_1 + 1. \end{aligned} \tag{6.1}$$

All six values are natural. These equations express exact remainders, including the cases $y = 0$, $S = 0$, or $S' = 0$. Their actual moduli are always positive. Let B_β denote the sum of their four residual squares. The one local relation is the polynomial equation

$$\mathcal{P}(i, \boldsymbol{a}, S, q_0, s_0, S', q_1, s_1) = B_\beta + \prod_{j=0}^{11} D_j = 0. \tag{6.2}$$

It has exactly ten local witnesses: four base parameters and six beta coordinates. The outer parameters are N, x, y, M_0.

The endpoints assert $\beta(x, y, 0) = M_0 + 2$ and $\beta(x, y, N) = 1$, consistently with $M = M_0 + 1$. CRT supplies a beta encoding for every finite accumulator: choose y larger than all entries and divisible by all positive integers up to N, then use the pairwise coprime moduli $1 + (i + 1)y$. This choice of y belongs to the existence proof; no factorial-divisibility constraint on y is added to the polynomial.

7 Bounded-family elimination with two bounds

Write $\mathcal{P} = e - f$, where e, f are nonnegative-coefficient polynomials represented by two natural arithmetic graphs. For example, $(e - f)(g - h) = (eg + fh) - (eh + fg)$. These graphs are

retained for the majorant argument even though later arithmetic predicates are compiled with signed intermediate values.

Let the ten bounds be

$$\boldsymbol{b} = (N, N, N, N, U, U, U, U, U, U).$$

For fixed outer parameters set

$$\begin{aligned} &L = e(N, \boldsymbol{b}) + f(N, \boldsymbol{b}) + N, \\ &T = L!, \qquad d_i = 1 + (i+1)T, \qquad B = \prod_{i<N} d_i. \end{aligned} \tag{7.1}$$

The precise fixed-parameter arguments of e, f are those in the literal source graph. Their occurrence is not an additional tuple of unknowns.

The bounded family is equivalent to a finite existential certificate with encoded columns $a_0, \ldots, a_9$ and an encoded index z:

$$\begin{aligned} &B \mid (b_j+1)!\binom{a_j}{b_j+1} && (0 \le j < 10), \\ &B \mid 1 + (z+1)T, \\ &e(z, \boldsymbol{a}) \equiv f(z, \boldsymbol{a}) \pmod{B}. \end{aligned} \tag{7.2}$$

Here the family means $\forall i < N\ \exists \boldsymbol{w} \le \boldsymbol{b}\ e(i, \boldsymbol{w}) = f(i, \boldsymbol{w})$.

Lemma 7.1 (Soundness of the bounded certificate)**.** *Conditions* (7.2) *imply the bounded family.*

Proof. Fix $i < N$, and choose a prime $p \mid d_i$. Since $d_i = 1 + (i+1)L!$, the prime cannot divide $L!$, so $p > L$. In particular $p > N$, and T is invertible modulo p. The encoded-index condition gives $z \equiv i \pmod{p}$.

For a column $a = a_j$, if $a < b_j + 1$, take $w_j = a$. Otherwise the product $(b_j+1)!\binom{a}{b_j+1} = a(a-1)\cdots(a-b_j)$ is divisible by p, so some factor gives $a \equiv w_j \pmod{p}$ with $0 \le w_j \le b_j$. This argument does not require the prime to exceed every coordinate bound. The encoded evaluation congruence yields $e(i, \boldsymbol{w}) \equiv f(i, \boldsymbol{w}) \pmod{p}$. Nonnegative-coefficient monotonicity bounds both values by their upper evaluations, hence by $L < p$. They are equal as integers. □

Lemma 7.2 (Completeness of the bounded certificate)**.** *If the bounded family holds, encoded columns and an index satisfying* (7.2) *exist.*

Proof. The finitely many moduli d_i are pairwise coprime. Indeed a common prime divisor of d_i, d_j is coprime to T and divides $|i-j|$, which is at most $N \le L$ and hence divides $L!$, a contradiction. Choose natural CRT representatives with $a_j \equiv w_{ij} \pmod{d_i}$ and $z \equiv i \pmod{d_i}$. Each falling factorial vanishes modulo every d_i: if its upper argument is too small the integer binomial coefficient is zero, and otherwise one factor has the required residue. The evaluation and index conditions hold modulo each factor and therefore modulo B. For $N = 0$, the empty product is one and the certificate is vacuous; the separate endpoints still reject that case. □

To choose U in a completeness construction, first fix the proof, N, M, the accumulator and its beta encoding. Choose U above all six beta coordinates in the finitely many rows. The ten-column family then holds with its mixed bounds. Recompute T, B for those bounds; one must not simply enlarge the bounds in a preexisting CRT certificate.

Only two different bound factorials are needed. The U-columns share $(U+1)!$, and the four N-columns use

$$(N+1)! = (N+1)N!. \tag{7.3}$$

The factor $N!$ is already required to construct B. Thus moving to mixed bounds adds no factorial certificate.

8 Final refinement: eight columns and no bound factorials

The ten-column construction is a checked intermediate, yielding the 9,476-integer quartic recorded in the supplementary comparison. The released equation adds two further exact reductions. They remove eleven power certificates from that intermediate and do not change the proof calculus or the final degree.

8.1 Bound beta remainders by the beta scale

In completeness we may choose the beta scale y larger than every accumulator entry before choosing its code x. Thus it is legitimate to require both adjacent remainders to satisfy $0 \le S, S' \le y$. For every $i \ge 0$,

$$y < 1 + (i+1)y \quad \text{and} \quad y < 1 + (i+2)y.$$

These inequalities also hold at $y = 0$. Consequently the two equations

$$x = q_0(1 + (i+1)y) + S, \qquad x = q_1(1 + (i+2)y) + S' \tag{8.1}$$

already express the exact two remainders when $S, S' \le y$. The two remainder-slack columns and their defining rows can be deleted. The final local bound vector, in the stored coordinate order, is

$$\boldsymbol{b} = (N, N, N, N, y, U, y, U), \tag{8.2}$$

for $(a, b, c, d, S, q_0, S', q_1)$. The local polynomial is obtained from (6.2) by replacing its four beta residuals by the two in (8.1); the branch factors (4.1) are unchanged.

For soundness, the decoded bounds supply the missing strict remainder conditions. For completeness, choose y as stated, choose x by CRT, and then choose U above the finitely many quotient values. No bound on x or U is used to choose M. All witness choices therefore retain their correct dependency order.

8.2 Cancel the bound factorials using coprimality

Enlarge the factorial argument to

$$L = e(N, \boldsymbol{b}) + f(N, \boldsymbol{b}) + N + U + y + 1, \qquad T = L!. \tag{8.3}$$

Every entry b_j of (8.2) satisfies $b_j + 1 \le L$. Hence $F_j = (b_j + 1)!$ divides T. For $B = \prod_{i<N}(1 + (i+1)T)$, each factor is one modulo F_j, so

$$B \equiv 1 \pmod{F_j}, \qquad \gcd(B, F_j) = 1. \tag{8.4}$$

It follows that

$$B \mid F_j \binom{a_j}{b_j + 1} \quad \longleftrightarrow \quad B \mid \binom{a_j}{b_j + 1}. \tag{8.5}$$

Thus the final coordinate-range certificates contain just eight binomial divisibilities, with no separate factorial value for any of their bounds. The factorials $L!$ and $N!$ remain because they are part of the CRT product construction; $(U+1)!$ disappears entirely.

Theorem 8.1 (The final bounded-family interface)**.** *For the eight-column row, fixed outer parameters and bounds* (8.2)*, take T from* (8.3)*. The bounded local family is equivalent to existence of encoded columns and an encoded index satisfying*

$$\begin{aligned} &B \mid \binom{a_j}{b_j + 1} && (0 \le j < 8), \\ &B \mid 1 + (z+1)T, \qquad e(z, \boldsymbol{a}) \equiv f(z, \boldsymbol{a}) \pmod{B}. \end{aligned} \tag{8.6}$$

Proof. The proof of the bounded-certificate lemmas remains valid when their factorial argument is enlarged: it is still at least the two upper evaluations and N, and it still ensures pairwise coprime moduli. Apply those lemmas to the eight-coordinate source and use (8.4)–(8.5) on each column. At $N = 0$, $B = 1$, so the same argument applies to the empty product; the endpoint contradiction still excludes that case from the complete equation. □

Two deleted binomial certificates save six powers. Removing the one bound-factorial certificate saves five more. The final complete ledger therefore has forty power certificates, eleven binomial certificates, twenty-seven division/remainder certificates, and two factorial certificates. The slightly larger factorial argument itself is a fixed polynomial expression and does not add an arithmetic representation certificate.

9 Removing all nonpolynomial notation

Equations (5.2), (8.3) and (8.6) are an explanation of the construction, not the final Diophantine language. The delivered compiler replaces every power, factorial, binomial and remainder by fixed finite polynomial certificates. It retains the baseline's pinned arithmetic interfaces [4]. The principal identities are stated here so that the elimination is reviewable.

9.1 The finite product

With $T \geq 1$, let

$$Q = (1 + (N+1)T)^N + 2.$$

Then the CRT product is represented by

$$B = \left(T^N N! \binom{Q+N}{N} \right) \bmod (TQ - 1). \tag{9.1}$$

The modulus is positive, including $N = 0$, because $Q \geq 3$. For completeness, this identity has a short direct proof. One has

$$T^N N! \binom{Q+N}{N} = \prod_{j=1}^{N} (TQ + jT) \equiv \prod_{j=1}^{N} (1 + jT) = B \pmod{TQ - 1}.$$

If $A = (1 + (N+1)T)^N$, then $B \leq (1 + NT)^N \leq A < TQ - 1$, because $Q = A + 2$ and $T \geq 1$. Thus the indicated remainder is exactly B. The empty product case $N = 0$ gives one on both sides.

9.2 Binomial coefficients and factorials

For $n, k \in \mathbb{N}$, put $q = 2^n + 1$. The base-q digit identity gives

$$\binom{n}{k} = \left\lfloor \frac{(q+1)^n}{q^k} \right\rfloor \bmod q. \tag{9.2}$$

Every binomial coefficient in the expansion of $(q+1)^n$ is below q, so there is no carry. If $k > n$, the extracted digit is zero. For $r = (n+2)^{n+2}$ and $C = \binom{r+n}{n}$, the pinned factorial identity is

$$n! = \left\lfloor \frac{(r+1)^n + C - 1}{C} \right\rfloor. \tag{9.3}$$

It includes $n = 0$; C is always positive. To see the identity for $n \geq 2$, write

$$\frac{(r+1)^n}{C} = n! \prod_{j=1}^{n} \frac{r+1}{r+j}.$$

The product is at most one and is greater than $1 - n(n-1)/(2r)$. Since $n!n(n-1)/2 < r$, the ratio lies in $(n! - 1, n!]$, so its ceiling is $n!$. Integer ceiling division is exactly the right side of (9.3). The cases zero and one are immediate. The predecessor $C - 1$ and all integer quotients are represented by polynomial rows.

9.3 Pell and exponentiation certificates

Write

$$\mathcal{N}(a, x, y) = x^2 + y^2 - a^2y^2 - 1.$$

For $a > 1$, the graph of the nonnegative Pell coordinates at index n is represented by $n \leq y$ and the disjunction of $(x, y) = (1, 0)$ with existence of natural u, w, s, t, b satisfying

$$\begin{gathered} \mathcal{N}(a, x, y) = \mathcal{N}(a, u, w) = \mathcal{N}(b, s, t) = 0, \\ b > 1, \quad w > 0, \quad y^2 \mid w, \\ b \equiv 1 \pmod{4y}, \quad b \equiv a \pmod{u}, \quad s \equiv x \pmod{u}, \quad t \equiv n \pmod{4y}. \end{gathered} \tag{9.4}$$

These are the finite Matiyasevich conditions used by the frozen baseline. The compiler includes their zero-index branch. It does not silently assume that all moduli in an inactive branch are nonzero.

The natural power graph has the explicit cases

$$(k = 0 \wedge m = 1) \ \vee \ (k > 0 \wedge [(n = 0 \wedge m = 0) \vee (n > 0 \wedge \mathcal{P}(m, n, k))]).$$

The nonzero case $\mathcal{P}$ uses natural w, a, t, z, x, y, the Pell graph at (a, k, x, y), and

$$\begin{aligned} &x \equiv y(a \dot{-} n) + m \pmod{t}, \qquad 2an = t + n^2 + 1, \\ &m < t, \quad n \leq w, \quad k \leq w, \qquad \mathcal{N}(w + 1, a, wz) = 0. \end{aligned} \tag{9.5}$$

Truncated subtraction is given its two-branch polynomial definition. Natural slack variables express inequalities. Sums of squares implement conjunction and products of nonnegative branch polynomials implement disjunction. The frozen arithmetic proofs justify these unbounded interfaces; bounded numerical examples are not used in their place.

Arithmetic certificate	Direct baseline	New projected source
Local witness columns	29	8
Natural power	108	40
Indexed Pell	108	40
Binomial	33	11
Factorial	3	2
Division/remainder	72	27

Table 2: Counts in the complete arithmetic-call ledger, before final conversion to one quartic over integers.

10 Signed arithmetic and congruence quotients

The local majorant still uses separate nonnegative expressions e, f, but there is no reason to require every intermediate arithmetic wire in the final integer polynomial to be natural. This distinction permits a more economical final compiler.

Lemma 10.1 (A single signed congruence quotient)**.** *For natural a, b, m, including $m = 0$,*

$$\exists u, v \in \mathbb{N}\ (a + mu = b + mv) \quad \longleftrightarrow \quad \exists q \in \mathbb{Z}\ (a - b + mq = 0).$$

Proof. Use $q = u - v$. Conversely choose $u = \max(q, 0)$, $v = \max(-q, 0)$. If $m = 0$, both sides say $a = b$. Thus the replacement remains valid inside inactive branches and does not depend on a missing nonzero-modulus hypothesis. □

Lemma 10.2 (Division quotients need no independent sign guard)**.** *Suppose $n, r, s \in \mathbb{N}$, $q \in \mathbb{Z}$, and*

$$n = qd + r, \qquad d = r + s + 1.$$

Then $d > 0$, $0 \le r < d$, and $q \ge 0$.

Proof. The first two inequalities follow from the second equation. If $q \le -1$, then $n \le -d + r = -s - 1 < 0$, a contradiction. □

This inference applies to unconditional division graphs whose arguments have already been proved natural. It does not authorize discarding arbitrary inequality slacks or positivity guards in conditional branches. The exact released set of unguarded quotient inputs is recorded and checked against the call ledger.

With signed arithmetic, the square of a residual is computed directly: one multiplication for $(a - b)^2$, after an affine subtraction. A pair representation would compute $a^2 + b^2 - 2ab$ using three products. Similarly, a product of two signed values needs one product, whereas multiplying their positive and negative parts uses four. These are exact integer identities, so they remain valid at every assignment, not only on the zero set. Multiplication-gate values may be negative; their sign is immaterial when the gate equations are squared at the final stage.

11 One ordinary quartic over unrestricted integers

Lemma 11.1 (Signed graph flattening)**.** *Consider a finite acyclic graph over integer constants, semantic inputs, addition, subtraction and multiplication, with specified output equalities. Introduce one integer wire for each genuinely nonlinear multiplication and substitute affine wires, optionally retaining selected affine wires to limit expression growth. Then every retained graph equation is quadratic. Their simultaneous zero set is equivalent to the original output equalities, with unique reconstructed graph values for fixed semantic inputs.*

Proof. Proceed in topological order. Constants and semantic inputs are affine. An addition or subtraction of affine forms is affine. A product of two current affine forms is quadratic; if it remains affine, substitute it, and otherwise introduce a wire with that quadratic defining row. A retained affine wire simply receives its affine defining row, with the same topological reconstruction argument. Consequently all later represented nodes are again affine forms in the semantic inputs and retained wires. Output equalities give affine rows. An actual graph evaluation supplies every wire. Conversely, each defining row forces the corresponding gate value, so induction reconstructs the whole graph and its outputs over $\mathbb{Z}$. □

Only semantic inputs whose domain is genuinely natural need a natural number guard. For each such input x, add three unrestricted integer witnesses and the quadratic equation

$$x = u^2 + v^2 + w(w+1). \tag{11.1}$$

The right side is nonnegative for every integer w. Conversely every natural x has such a representation: $4x+1$ is not of the excluded form in the three-square theorem; its three-square representation has exactly one odd square. Dividing $4x+1 = (2u)^2 + (2v)^2 + (2w+1)^2$ by four proves the claim. This standard conversion is also stated explicitly by Sun [5].

Congruence quotients covered by Lemma 10.1 have unrestricted domain. Other unguarded quotient variables have their natural domain derived by the preceding division lemma, with its hypotheses checked. The remaining semantic inputs are guarded using (11.1). There is no guard for every internal multiplication wire: the signed graph theorem already accounts for them.

The final polynomial is the sum of the squares of all graph rows, output rows and semantic guards. Over $\mathbb{Z}$, a sum of integer squares vanishes if and only if every row vanishes. It has degree at most four. The expanded delivered coefficient list contains nonzero monomials of degree four, which certifies exact degree. In particular, the square of a genuine natural-number guard contributes a fourth-power term; the leading homogeneous sum of squares cannot cancel identically over the reals.

The released dimension count is explicit:

Coordinate role	Count
Semantic inputs	1,268
Nonlinear multiplication wires	3,198
Retained affine wires	31
Natural-number representation witnesses	3,120
Total unrestricted integer coordinates	7,617

Of the semantic inputs, 201 are signed congruence quotients and 27 are Euclidean quotients with derived nonnegativity. The remaining 1,040 inputs receive three witnesses each, giving $3 \cdot 1040 = 3120$. The implementation retains 31 affine wires when substitution would exceed its fixed sixteen-term expression threshold. Their equations are linear and are counted explicitly. Together with 110 output constraints and 1,040 natural guards, the graph gives $3198 + 31 + 110 + 1040 = 4379$ squared residuals.

12 Proof of the consistency equivalence

Proof of Theorem 1.2. Suppose the integer polynomial has a zero. Its sum of squares yields every quadratic row. The natural guards and derived quotient signs give the precise semantic domains. Signed graph reconstruction recovers the finite arithmetic certificates. The pinned Pell, power, binomial, factorial and remainder theorems therefore give the stated arithmetic values, without any remaining uninterpreted arithmetic function.

Theorem 8.1 recovers, for each $i < N$, eight natural local witnesses bounded by $\boldsymbol{b}$. The two beta equations and decoded remainder bounds identify adjacent values of one accumulator. Since each branch polynomial is nonnegative, one branch vanishes and supplies a valid C222 rule with the correct update. Projection restores all omitted constructor wires by their polynomial values. The four base parameters are at most N, so $H(N)$ bounds every label. The endpoints, the binary height equation and the accumulator yield the target balance and hence $\mathcal{C}_{222}(1)$. The two inherited logical calibrations (2.3) yield a finite ZFC refutation.

Conversely, suppose there is a finite ZFC refutation. The same calibration gives a finite C222 derivation. Expand its proof tree, pad it by truth rules as in Theorem 5.1, and choose N before choosing the base. Define the accumulator, choose a beta encoding and then a large enough U. Bounded-certificate completeness supplies the encoded index and columns. The unbounded arithmetic representation theorems supply all their finite natural witnesses. Replace congruence quotient pairs by their signed differences. Evaluate the signed graph, and represent every required natural input by (11.1). All final residuals vanish, so the polynomial is zero.

We have proved in ordinary mathematics $\exists \boldsymbol{x} \in \mathbb{Z}^{7,617} P_{\mathrm{direct}}(\boldsymbol{x}) = 0 \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}$. The finite combinatorics, CRT, Pell arithmetic and natural-number arguments are formalizable in ZFC; the inherited logical calibration is likewise an ordinary ZFC argument. Negating the equivalence gives (1.2). No step assumes that ZFC is consistent. Exact degree follows from the preceding section. □

The proof has two clearly identified layers. The new contribution proves the projection, padding, bounded-family specialization, factorial cancellation and signed compilation transformations. The arithmetic representation library and the C222 logical calibration are imported results with frozen source files and explicit theorem names. The finite verification scripts certify literal bindings and algebraic identities. They do not independently establish every unbounded imported theorem, or produce a new kernel proof of the complete consistency statement.

13 Verification and reproducibility

The archive contains the full expanded polynomial, its quadratic rows, the final signed arithmetic graph, the projected natural row graph, the fixed Horn-rule data, builders, independent checks and receipts. Source hashes pin the reviewed input archive to repository commit `04a1d79222d55cabe4ba52897863fc33baa8c268`. The inherited logical files retain their earlier source pin, `9a33342a03d19cf5b061b2083c6b5d3f27f752ed`, as recorded in the separate provenance manifest. The native coefficient ports and Turing-machine transition table are not needed for this direct equation.

The checks cover the following mathematically different obligations:

1. Exact projection of each of the twelve branches, including all constructor definitions, premise/conclusion labels and the retained bundle side condition.
2. The two occurrences of the local natural source graph: upper evaluation at $(N, \boldsymbol{b})$, and encoded evaluation at the encoded index and eight encoded columns.
3. The literal mixed bounds, the endpoint equations, the diagonal label majorant, and the enlarged factorial argument that permits cancellation of every bound factorial.
4. Signed arithmetic identities, congruence replacement and the precise semantic-domain list. Inferred nonnegativity is attached to actual unconditional division interfaces.
5. Reconstruction of the affine forms and quadratic graph equations, followed by exact expansion and collection of their sum of squares.
6. Variable count and liveness, actual degree, monomial count, coefficient height, and deterministic regeneration of the final artifacts. Mutation controls ensure that selected corrupted inputs or artifacts are rejected.

The accompanying verification notes state the exact commands run and their results. The finite checks establish literal structural identities; they are not used as a substitute for proofs of the unbounded relations. No enumeration of all integer tuples, consistency test, proof search for a ZFC contradiction, or simulation of K215 was performed.

14 What improved, and what did not

The principal improvement is a reduction of 20,299 unrestricted integer variables in a completely materialized direct quartic. The final degree remains four. Coefficient height is a tradeoff: the new maximum absolute coefficient is 8 974 652 400 619 654 748 160, versus 320307372009918 in the baseline; it is not a coefficient-height improvement. The improvement arises before and during compilation, not from treating a long list of hidden variables as notation. The coefficient table is supplied in full.

The most reusable mathematical idea is this ordering of operations: first substitute deterministic data separately in each finite rule branch; next use a neutral rule to couple proof length to a parameter bound; only then encode the bounded family; finally flatten signed arithmetic. Padding removes a potential circular bound while avoiding three extra label-slack columns. Bounding remainders by the beta scale removes two additional columns; factorial cancellation removes one entire arithmetic certificate.

This equation does not improve the repository's fewest-variable constructions. Its nine-positive-variable equations and ten-integer equations occupy a different part of the tradeoff. Sun's ten-unknown theorem is a current general result [5]; no claim that ten is globally minimal follows from it. Nor does our degree-four construction prove that a still simpler consistency equation cannot exist.

We investigated a proposed smaller-degree modification of the thirteen-integer radical construction. Replacing its scale $Y = qv$ by a constant multiple of v appeared to lower polynomial degree, but destroyed the universal coprimality $\gcd(H, q) = 1$ used by its CRT recovery. That candidate is not included as a valid equation. It illustrates why degree counts alone cannot establish a Diophantine reduction.

Direct relation-algebra encodings are another worthwhile direction. Formisano, Gavazzi and Omodeo discuss a direct encoding of NBG provability using an equational relation calculus [6]. The present work uses the already calibrated numerical C222 source instead of claiming that their separate axiom-translation project is complete.

15 Further research questions

1. **Optimize column count before all arithmetic encoding.** Can one eliminate one of the four beta remainder/quotient columns without introducing an equally expensive bound or division certificate elsewhere? A candidate must preserve the shared accumulator and both endpoint semantics.
2. **Share exponentiation certificates.** The padded source still uses 40 power certificates. Which repeated base/exponent pairs can be shared extensionally, with all zero-base and zero-exponent branches preserved? An exact call-graph proof is needed; syntactic similarity alone is insufficient.
3. **Represent the bounded column test more cheaply.** Each falling-factorial divisibility condition is currently compiled through a binomial coefficient. Is there a lower-cost fixed polynomial interface when several columns share the same bound?
4. **Improve the proof calculus itself.** A different equiconsistent finite single-conclusion calculus with fewer effective rule parameters could reduce the expensive local dimension further. It must include explicit numerical rules and two-way syntactic calibration to ZFC refutability.
5. **Search the coefficient/variable frontier.** Affine elimination can increase coefficient height and monomial count. Retaining selected affine wires may yield a useful family of smaller-coefficient quartics; those tradeoffs should be measured on literal polynomials rather than estimated from source syntax.

6. **Transfer the new source into low-variable compression.** The direct quartic can in principle feed a universal compression. A worthwhile result must materialize its actual coefficients and compare the resulting degree, numeral size and arithmetic graph against the existing native constructions. An abstract appeal to MRDP is not a completed improvement.
7. **Formalize the new transformations.** The branch projection and padding lemmas are compact targets for proof-assistant verification. A complete formal endpoint must also identify the emitted coefficient list with its polynomial and explicitly internalize the consistency equivalence; an external equivalence alone is a different statement.

A Complete polynomial rule schemata

Use the constructors in (2.1). The eight logical axiom labels are

$$\begin{aligned}
\rho_1(a,b) &= \mathsf{X}(a,\mathsf{E}(a,b)),\\
\rho_2(a,b,c) &= \mathsf{I}(\mathsf{A}(a,\mathsf{I}(b,c)),\mathsf{I}(\mathsf{A}(a,b),\mathsf{A}(a,c))),\\
\rho_3(a,b,c) &= \mathsf{I}(\mathsf{E}(a,b),\mathsf{I}(\mathsf{R}(c,a),\mathsf{R}(c,b))),\\
\rho_4(a,b,c) &= \mathsf{I}(\mathsf{A}(a,\mathsf{A}(b,c)),\mathsf{A}(b,\mathsf{A}(a,c))),\\
\rho_5(a,b,c) &= \mathsf{I}(\mathsf{E}(c,a),\mathsf{I}(\mathsf{R}(c,b),\mathsf{R}(a,b))),\\
\rho_6(a,b) &= \mathsf{I}(\mathsf{X}(a,\mathsf{A}(a,b)),b),\\
\rho_G(a,b,c) &= \mathsf{I}(\mathsf{E}(c,a),\mathsf{I}(\mathsf{E}(c,b),\mathsf{A}(\mathsf{R}(a,b),\mathsf{E}(a,b)))),\\
\rho_8(a,b,c,d) &= \mathsf{I}(\mathsf{I}(\mathsf{I}(c,d),a),\mathsf{I}(\mathsf{I}(a,c),\mathsf{I}(b,c))).
\end{aligned}$$

For the bundle define, as nested polynomial abbreviations,

$$\begin{aligned}
S(a,b,k) &= \mathsf{A}(k,\mathsf{I}(\mathsf{R}(k,a),\mathsf{R}(k,b))),\\
\mathrm{Req}(b,k,m) &= \mathsf{A}(b,\mathsf{I}(m,\mathsf{N}(\mathsf{A}(b,\mathsf{I}(m,\mathsf{N}(\mathsf{R}(b,k))))))),\\
\mathrm{Clos}(a,b,k,m) &= \mathsf{A}\big(b,\mathsf{I}(\mathsf{R}(b,k),\mathsf{N}(\mathsf{A}(a,\mathsf{I}(m,\mathsf{N}(\mathsf{A}(a,\mathsf{I}(S(a,b,k),\mathrm{Req}(b,k,m)))))))))\big),\\
\mathrm{NKS}(a,b,c) &= \mathsf{A}(c,\mathsf{I}(\mathrm{Clos}(a,b,c,c),\mathsf{N}(\mathrm{Req}(b,c,c)))),\\
\mathrm{VR}(a,b,c) &= \mathsf{I}(\mathsf{R}(a,b),\mathsf{A}(c,\mathsf{R}(a,b))),\\
\mathrm{Bundle}(a,b,c) &= \mathsf{N}(\mathsf{I}(\mathrm{VR}(a,b,c),\mathrm{NKS}(a,b,c))).
\end{aligned}$$

The literal frozen source and the independent branch checker are the authoritative finite binding of these expressions to the emitted graph.

Rule	Premise 1	Premise 2	Conclusion
Truth	0	0	0
Each of $\rho_1,\dots,\rho_6,\rho_G,\rho_8$	0	0	its displayed label
Bundle	0	0	$\mathrm{Bundle}(a,b,c)$
Modus ponens	a	$\mathsf{I}(a,b)$	b
Generalization	b	0	$\mathsf{A}(a,b)$

Only the bundle requires (2.2); all parameters range over $\mathbb{N}$. This is a finite numerical calculus whose formulas happen to encode logical expressions. The final polynomial does not quantify over formulas or invoke a parser.

B Artifact conventions and integration

The top-level README specifies the supported build and verification commands, input provenance, complete file roles and the recommended equation. The generated metrics file used in this paper is produced from the same released artifact checked by the verifier. A release hash manifest covers the final files.

For integration into ProveIt, preserve this contribution as an additional package under the Diophantine-consistency research material, or retain the archive in `docs/incoming`. Its direct compiler does not replace the low-variable native branch or the inherited formalization. The frozen source files retain their original licenses and notices; new files are explicitly identified. No remote repository file was changed as part of preparing this package.

References

[1] ProveIt research archive, *Direct and centered Diophantine consistency equations*, Round12, source package labeled 24 October 2026. Pinned in repository commit `04a1d79222d55cabe4ba52897863fc33baa8c268`, https://github.com/VladimirReshetnikov/ProveIt/blob/04a1d79222d55cabe4ba52897863fc33baa8c268/docs/incoming/ConZFC_Direct_Centered_Round12_2026-10-24.zip.

[2] ProveIt, numerical C222 calculus and logical endpoint, `Code254.lean`, `KCode222.lean`, `KEndpoint.lean`; frozen source snapshots and their original repository paths are included in this package's provenance records. See also https://github.com/VladimirReshetnikov/ProveIt/tree/9a33342a03d19cf5b061b2083c6b5d3f27f752ed/Computability/BusyBeaver/Lean/BusyBeaver/ZFC.

[3] ProveIt, `ConstructibleChoice.lean`, especially `zf_refutation_iff_zfc_refutation`; frozen source snapshot in the supplied provenance material. The mathematical input is Goedel's constructible-universe equiconsistency argument.

[4] ProveIt, `PellRepresentation.lean`, `PowerRepresentation.lean`, `FactorialRepresentation.lean`, `BoundedForallRepresentation.lean`, and `BoundedForallCertificate.lean`; pinned source snapshots inherited from Round12, retained with this contribution.

[5] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2026, https://arxiv.org/html/2609.01594v2.

[6] A. Formisano, I. Gavazzi and E. G. Omodeo, *Diophantine Modeling of Provability in Algebraic Logic*, CILC 2025, CEUR Workshop Proceedings, vol. 4003, https://ceur-ws.org/Vol-4003/paper13.pdf.